LOKESH KUMAR (SONARA)

Software Engineering Researcher & Lead Systems Architect | Applied AI, Edge IoT & Low-Resource NLP

Location: Mithi (Tharparkar) & Tando Jam, Sindh, Pakistan | Nationality: Pakistani | DOB: 18/08/2003 | Passport: AA8310592
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Google Scholar: https://scholar.google.com/citations?user=NsmEvqMAAAAJ | ORCID iD: https://orcid.org/0009-0006-6510-6393
GitHub: github.com/lokeshsoni653-del | Hugging Face: huggingface.co/datasets/lokeshsonara/dhatki-agri-intents | IEEE: #99402233

PROFESSIONAL SUMMARY & RESEARCH OBJECTIVE

Final-year Software Engineering researcher at Sindh Agriculture University (SAU), Tando Jam, specializing in offline-first edge IoT telemetry, low-resource Indo-Aryan vernacular NLP (Dhatki), and scalable full-stack AI pipelines. Lead Systems Architect and proposal author for the $10,000 EPICS in IEEE grant initiative (Agri-Logistics IDAS) under the IEEE Karachi Section. Author of 4 peer-citable research manuscripts with CERN Zenodo DOIs, 4 flagship engineering platforms, and 3 data/software internships.

1. EDUCATION AND TRAINING

Bachelor of Science in Software Engineering (BS-SE) | CGPA: 3.00 / 4.00 (Peak Semester GPA: 3.65 / 4.00)
Information Technology Centre, Sindh Agriculture University (SAU), Tando Jam, Pakistan | 2022 - Expected Dec 2026

- Core Coursework: Data Structures & Algorithms, Discrete Structures, Artificial Intelligence, Operating Systems, Database Systems, Software Architecture & Design, Probability & Statistics, Computer Networks.
- Medium of Instruction: English (Certified by Official University Notification No. ENG 375; IELTS Academic Booked).

Higher Secondary School Certificate (HSC-II, Pre-Medical) - Grade A-1 (94.18% | 1036 / 1100 Marks)
Govt. Boys Degree College Mithi, District Tharparkar (BISE Mirpurkhas) | 2021 - 2023 (Secondary School SSC-II Science: Grade A, 2020)

2. FLAGSHIP APPLIED ENGINEERING PROJECTS

Agri-Logistics IDAS (Intelligent Decision & Advisory System) - Lead Systems Architect & Proposal Author
International EPICS in IEEE ($10,000 Grant Initiative) with IEEE Karachi Section & SPO | Portal: https://agri-idas.tech/ | 2026

- Engineered an offline-first edge IoT and vernacular logistics telemetry framework using ESP32 nodes, sub-GHz LoRa (868 MHz), and SQLite circular ring-buffers, achieving 99.4% packet integrity across 12-hour cellular blackouts and 38.6% water conservation in 48 C field trials.
- Demonstrated across 8 Lower Sindh transit corridors (1,313 OSRM points) that straight-line Haversine routing induces a 17.07% distance underestimation error (t=4.892, p=0.0017), preventing +8.84 L of unbudgeted diesel burn per trip.

World Bank Group (HQ) Sovereign Treasury AI & Financial Risk Engine - Lead Developer
Engineered for World Bank Group HQ Treasury Pioneers Program | Live: https://wbg-treasury.vercel.app/ | Next.js, React, Python | 2026

- Architected an automated sovereign debt risk scoring and yield-curve stress-testing platform processing multi-currency portfolio telemetry and generating executive risk audit dossiers.

NexusOmni (ContractIQ) - AI Legal Contract Risk Scoring Platform & Voice Automation Capstone FYP
Live Production Platform: https://nexus-omni-tdue.vercel.app/ & SAU Capstone FYP (Supervised by Madam Sohni Abbasi) | 2026

- Built an LLM-powered legal contract parser evaluating liability clauses and outputting 0-100 risk scores under 60 seconds; concurrently architected an OS-level desktop-to-mobile voice workflow automation engine for SAU Capstone FYP.

S-Jeewlani Jewellers Bullion E-Commerce Platform & Hyundai SmartDealership 360 Analytics Engine
Commercial Production Systems | Live: https://www.sjeewlanijewellers.com/ | Next.js 15, TypeScript, Leaflet GIS, Chart.js | 2026

- Deployed a live Sarafa bullion e-commerce platform with automated WhatsApp price pipelines, alongside a geospatial CRM & NLP customer sentiment triage platform for Hyundai Shahrah-e-Faisal, Karachi.

3. TECHNICAL & LANGUAGE SKILLS

- Programming & Web Stack: Python (Advanced), JavaScript (ES6+), TypeScript, SQL (PostgreSQL, MySQL, SQLite), C++, Next.js 15, React.js, Node.js, Streamlit, HTML5, CSS3, Tailwind CSS, RESTful APIs.
- AI, NLP & Data Engineering: Low-Resource NLP, Byte-Pair Encoding (BPE), Transformer Fine-Tuning (DistilBERT, mBERT, XLM-R), Speech Quantization, RAG Pipelines, Pandas, NumPy, NetworkX, Folium/Leaflet GIS.
- Edge IoT, Cloud & DevOps: ESP32/STM32 Microcontrollers, LoRa 868 MHz, MQTT, Git, GitHub, Docker, Google Cloud Platform (GCP), Vercel Global CDN, Linux/Bash.
- Languages (CEFR): Dhatki (C2 Native), Sindhi (C2 Native), Urdu (C2 Bilingual), English (C1 Proficient - Official MOI Certified).

4. RESEARCH PUBLICATIONS, PREPRINTS & OPEN-SOURCE AI DATASETS

[1] L. Kumar, "Benchmarking Intent Classification in Extremely Low-Resource Indo-Aryan Dialects: An Empirical Study on Dhatki Agricultural Telemetry"

CERN Zenodo Open Research Preprint | Nov 2026 | Permanent DOI: https://doi.org/10.5281/zenodo.23958454

- Constructed the first 1,000-utterance Dhatki agricultural intent corpus across 10 classes (published on Hugging Face); demonstrated that custom vernacular BPE tokenization with DistilBERT achieves 89.4% Micro-F1 at 14.2 ms edge CPU latency, outperforming mBERT by +7.8%.

[2] L. Kumar, "Resilient Edge-Computing Architectures for Precision Agritech Under Extreme Arid Conditions: The Thar Desert Telemetry Blueprint"

CERN Zenodo Technical Report | Nov 2026 | Permanent DOI: https://doi.org/10.5281/zenodo.23970079

- Formulated hardware-software co-design and polynomial thermal sensor calibration (restraining drift to +/-1.2% at 48 C vs +/-14.8% raw analog) paired with FAO-56 Penman-Monteith predawn micro-irrigation actuation.

[3] L. Kumar, "The Broken Ladder: Generative AI, Digital Feudalism, and the Structural Collapse of Entry-Level Tech Employment for Global South Graduates"

CERN Zenodo Socio-Technical Whitepaper | Nov 2026 | Permanent DOI: https://doi.org/10.5281/zenodo.23959079

- Analyzed multi-institutional graduate employment telemetry (n=420) across Pakistani universities, documenting a 42.8% contraction in entry-level coding roles and proposing a 4-pillar sovereign compute and verification curriculum framework.

[4] S. R. Suthar and L. Kumar, "Do States Eat Their Own? Predatory Debt, Digital Blackmail & Thar's Suicide Crisis: A Socio-Legal & Policy Intervention Brief"

CERN Zenodo Policy Brief (With SPO Sindh) | Nov 2026 | Permanent DOI: https://doi.org/10.5281/zenodo.23955391

- Investigated 27 socio-economic drivers and unlicensed mobile loan-app cyber extortion across Tharparkar and Umerkot, formulating FIA/PTA regulatory enforcement and grassroots psychological first-aid protocols.

[5] L. Kumar and B. S. Chowdhry, "Bridging the Digital Literacy Gap in Rural Agri-Logistics: A Trilingual, Context-Aware Intelligent Driver Assistance System"

SAU Tando Jam & MUET Jamshoro Joint Research Manuscript | Oct 2026 | PDF: https://agri-idas.tech/AgriLogistics_IDAS_IEEE_TechRxiv_Paper.pdf

- Evaluated trilingual (Dhatki, Sindhi, Urdu) acoustic intent pipelines achieving 0.0034 ms semantic routing and ISO 26262 compliant auditory voice synthesis in 747.92 ms for rural freight operators.

5. WORK & INTERNSHIP EXPERIENCE

Summer Intern - Volunteer Management & Welfare Logistics Department

Alkhidmat Foundation Pakistan (Ref: AKFP/VM/2883) | June 2026 - August 2026 (8 Weeks) | Verified: https://freeimage.host/i/nniprf1

- Spearheaded digital volunteer mobilization workflows, centralized database consolidation, and emergency relief logistics; awarded Official Completion Letter by Senior Executive Leadership.

Data, Operations & AI Intern

Pakistan Hindu Council | March 2026 - May 2026 (6 Weeks, Hybrid) | Verified: https://freeimage.host/i/CBVPY8l

- Restructured manual community data collection pipelines into automated Python reporting architectures, reducing operational reporting latency by 40% while enforcing anonymized database schemas.

Web Development Intern

Interns Pakistan | December 2024 - January 2025 (4 Weeks, Remote) | Verified: https://freeimage.host/i/C26XtQR

- Executed responsive front-end interface engineering across production web modules using HTML5, CSS3, and modern JavaScript.

6. HONOURS, LEADERSHIP (PORS) & VERIFIED CERTIFICATIONS

- Lead Proposal Author ($10,000 IEEE EPICS Grant): Principal student author for the IEEE Karachi Section & SPO community service grant.
- Active IEEE Leadership: IEEE Student Member (#99402233, IEEE Karachi Section) & IEEE Computer Society Member (2024 - Present); mentored junior SAU software engineering cohorts in Python, Git, and open-source research.
- Scholastic & National Distinctions: Scored 94.18% (Grade A-1) in HSC-II Board Exams; recipient of National Talent Award (Great Future Talent Olympiad, Dec 2025); competed among 525,000+ global contenders in QuizOff 2026 AI Competition.
- DeepLearning.AI (Prof. Andrew Ng): AI For Everyone (Credential ID: BTJWB77W6W2Q | coursera.org/verify/BTJWB77W6W2Q).
- IBM: Introduction to Software Engineering - SDLC & Architecture (Credential ID: E42GWWF22X5S | coursera.org/verify/E42GWWF22X5S).
- Google: Foundations of Cybersecurity (ID: 3NZWPQQJBMQ5) & Google Agile Essentials 3-Course Specialization (ID: GZW3MCSAGXNX).
- University of Michigan: Programming for Everybody - Python (Verified Coursera Credential).
- World Bank Group & Nestle Enterprise: Sustainable MSME Finance & ESG Compliance + NESTGEN 2026 (16 Verified Credentials in Industry 4.0, AI Ethics, Cloud SaaS & Circular Economy).